CBL: Case Based Learning

Medical Biochemistry and Molecular Biology Department

CBL: Case Based Learning

What?, Why?, How?

Why? To cross the gap between basic biomedical sciences and clinical practice

How? By analyzing clinical cases from biochemical point of view, you will be able to:

- Understand biochemical basis of diseases
- Interpret patient's investigations
- Understand the biochemical basis of therapy

Triad for Diagnosis

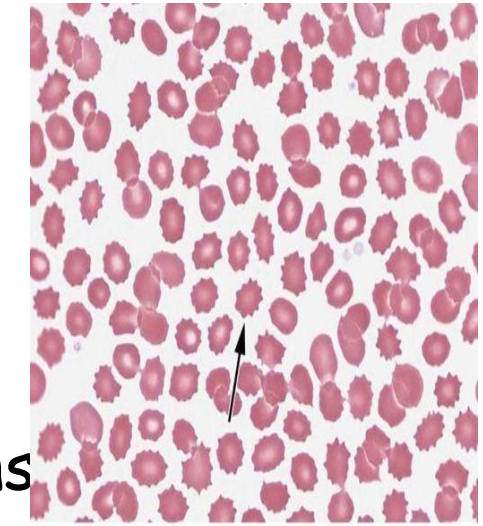
- History taking & Patients complaints (symptoms).
- Clinical examination (signs).
- Investigations (Laboratory, radiological & others).



Let’s study her case

A 7-day-old female was admitted to hospital suffering from pallor, lethargy and jaundice. History taking showed that she was jaundiced at day 1 of life but discharged without diagnosis and didn't receive any treatment. Mother and father were cousins from a small village in Sohag. Physical examination revealed hypotonia and mild splenomegaly.

Laboratory Investigations



- **CBC**

  Hb: 6 g/dL (N:12.7-18.3).
  Reticulocyte count: increased.
  Normocytic normochromic anemia.

- **Blood smear:** RBCs with thorny projections
- **S.total bilirubin:** 19 mg/dL (5–30 days: 0.0–1.8).
- **S.direct (conjugated) bilirubin:** 1 mg/dL (N:0.0–0.4).
- **S.indirect (unconjugated) bilirubin:** 18 mg/dl (N: 0-1.4).
- **Increased serum 2,3-BPG.**
- **RBC's enzyme assay:** Decreased **pyruvate kinase** activity (=15% of normal).

Patient's complaints

- Pallor.
- Lethargy.
- Jaundice.

Signs

- Hypotonia
- Splenomegaly
- Jaundice

Questions

1. What is your provisional diagnosis? Clues?
2. What is your differential diagnosis? Justify with evidence-based?
3. What are the required investigations to refine your diagnosis
4. What are the general characteristics of hemolytic anemia?.
5. What are the different types of hemolytic anemia?

6. What is the Diagnostic test for anemia caused by abnormalities in Hb?
7. What is the role of G6PD in RBCs?
8. Explain the presence of neurological symptoms PKD anemia.
9. Explain the occurrence of hyperbilirubinemia
10. Explain the biochemical basis of phototherapy in hyperbilirubinemia.

11. Mention different causes of deficiency anemias.

12. What are the unique characteristics of iron deficiency anemia?

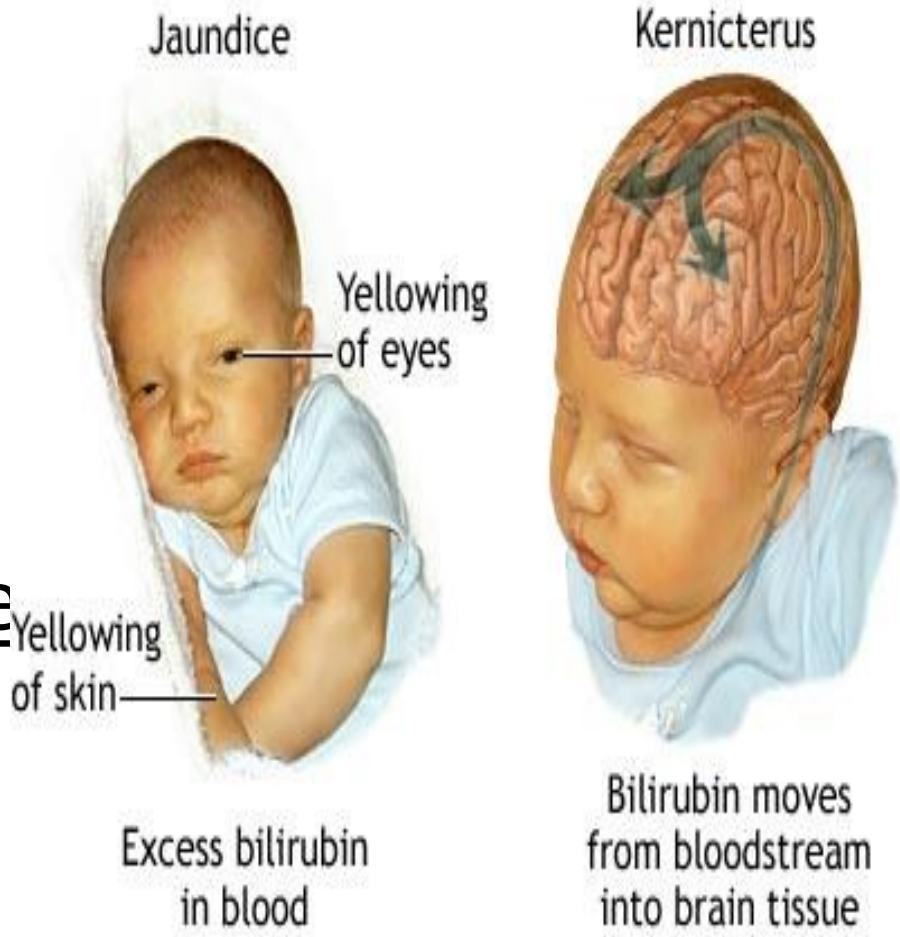
13. What's the daily iron requirement? What are iron sources?

14. What are the factors affecting iron absorption and comment on its regulation?

15. What are the specific treatment measures of iron deficiency anemia? .

1- What is your provisional diagnosis? Clues?

So, it is a case of <u>hemolytic anemia</u> <u>complicated with acute bilirubin</u> **<u>encephalopathy</u>** <u>(Kernicterus) , May be congenital.</u>



Clues for diagnosis of Hemolytic anemia:

1. Pallor and jaundice.
2. Normocytic anemia.
3. High bilirubin (mostly unconjugated).
4. ↑Reticulocytic count

Clues suggesting of encephalopathy:

1. Hypotonia.
2. Lethargy.

2- What is your Differential diagnosis? Justify with evidence-based

Hemolytic anemia:

- ***Normocytic normochromic with high reticulocytic count.***

➢**Hereditary spherocytosis.**

➢**Haemoglobinopathies (Abnormal hemoglobin):**

✓Sickle cell disease (**Hb S**)

✓Thalassemia

➢**Enzyme-deficient RBCs:**

✓G6PD deficiency

✓Pyruvate kinase deficiency

3-What is the required investigations to refine your diagnosis?

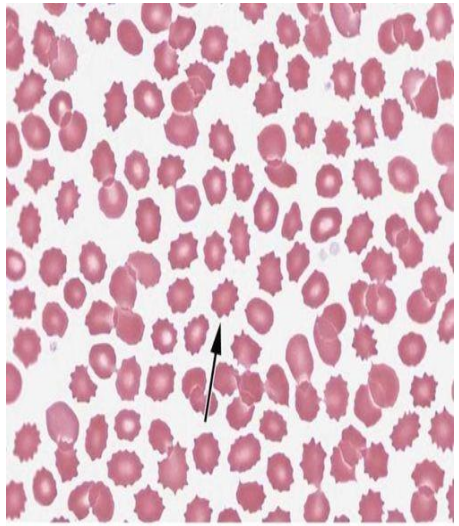
- 2,3-BPG (differentiate between different types of enzymatic deficiency).

- Enzyme assay (hexokinase-PK).

- Electrophoresis.

Remember your patient investigations…

Laboratory Investigations



- **CBC**

  Hb: 6 g/dL (N:12.7-18.3).

  Reticulocyte count: increased.

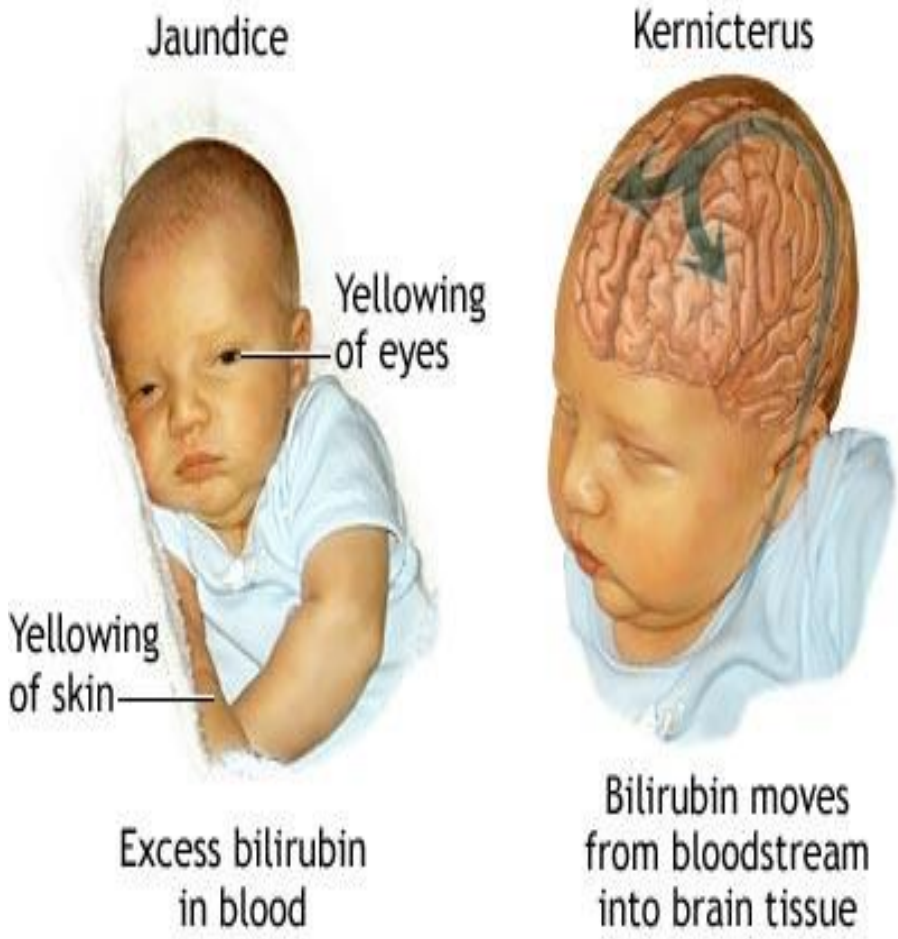
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- **Increased serum 2,3-BPG.**

- **RBC's enzyme assay:** Decreased **pyruvate kinase** activity (=15% of normal).

So , the diagnosis is Pyruvate kinase deficiency-hemolytic anemia complicated with acute bilirubin **encephalopathy** (Kernicterus).



Clues for diagnosis pyruvate kinase deficiency hemolytic anemia

Hemolytic anemia of pyruvate kinase deficiency is characterized by:

1. Being congenital disease, manifested early in life.
2. High 2,3-BPG (R-L cycle). Why?
3. Decreased PK enzyme activity: diagnostic.

Treatment plan

- **Treatment of anemia**

  Blood transfusion

  Folic acid and other vitamin B supplement

  Splenectomy.

- **Treatment of hyperbilirubinemia**

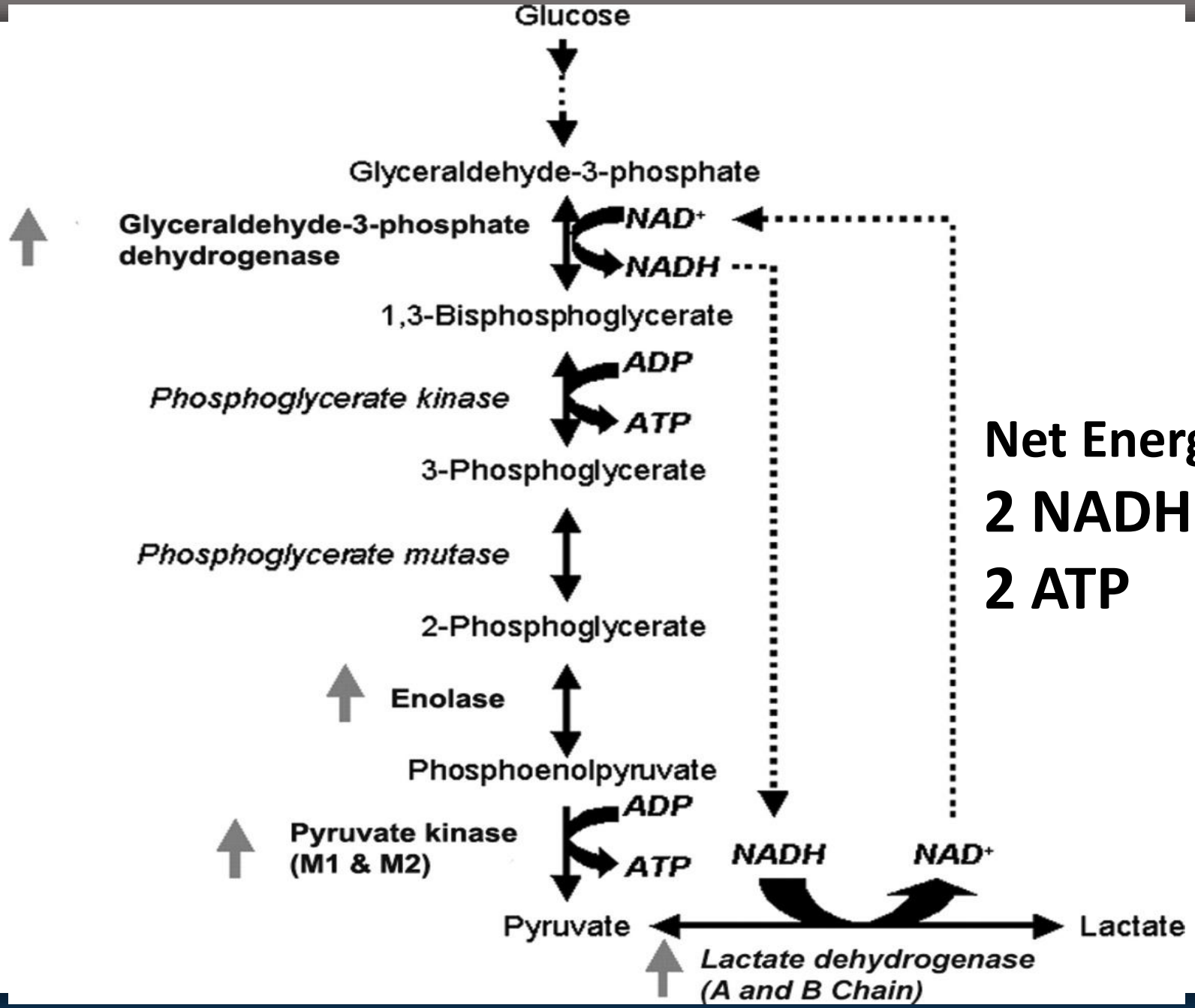
  Phototherapy

  Exchange transfusion

The affected pathway in this case is Glycolysis

- **Glycolysis:** glucose → Pyruvate (Aerobic) or Lactic (Anaerobic) with release of ATP.
- **Site:** Cytosol.
- **Two types:**

1. **Aerobic,** in tissues with mitochondria (for oxidative phosphorylation) e.g.: Neurons & heart.
2. **Anaerobic,** in tissues lack mitochondria (RBCs) or with low oxygen tension (e.g.: eye lens, cornea, retina, leukocytes, renal medulla, testis).



Net Energy yield:

2 NADH

2 ATP

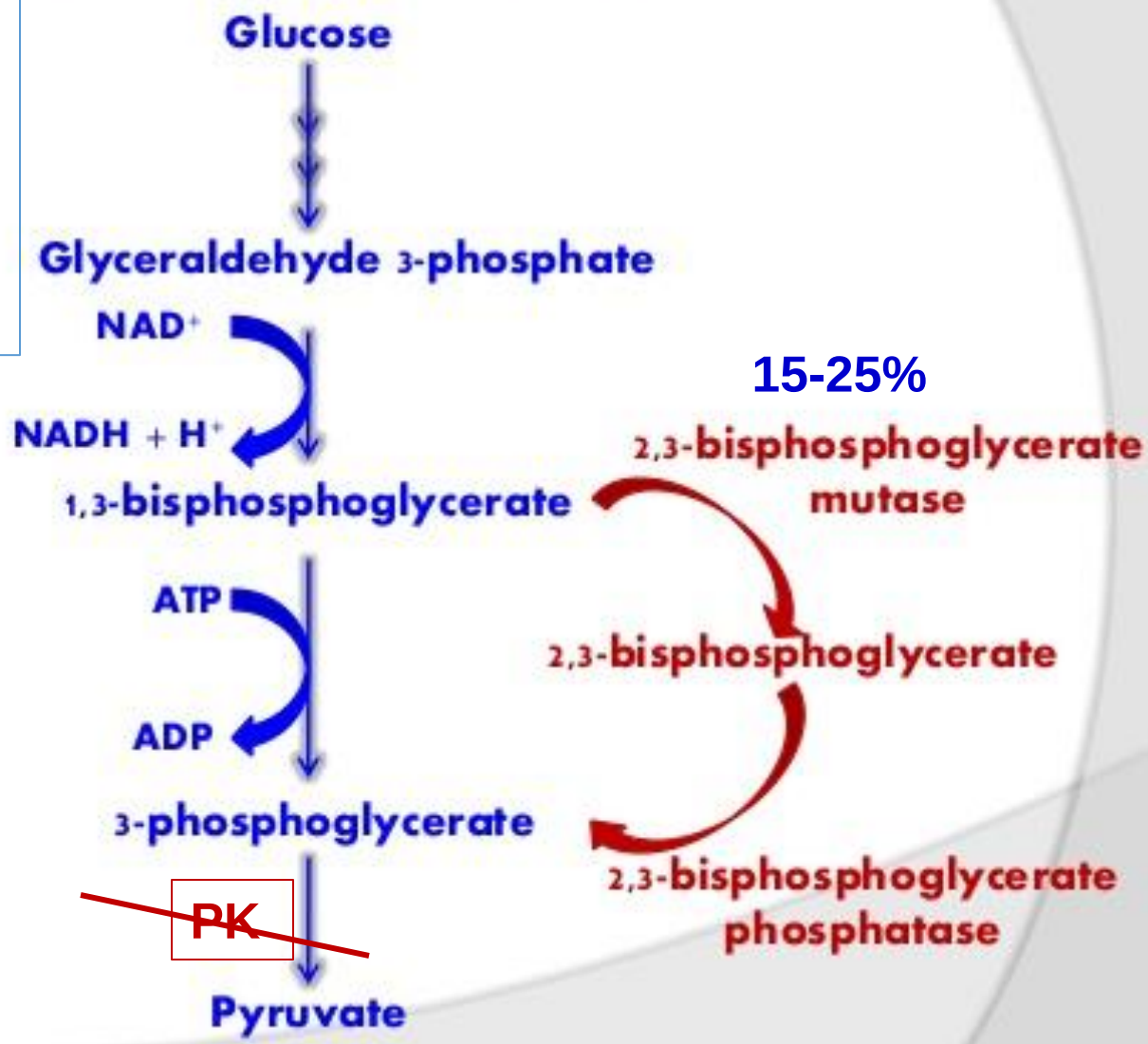
Pyruvate kinase as a key regulatory enzyme

- Pyruvate kinase is one of the key regulatory enzymes of glycolysis.
- Regulatory enzymes of glycolysis:

1. Hexokinase/Glucokinase.
2. Phosphofructokinase I.
3. Pyruvate kinase.

Rapaport-Leubering Cycle

1. What is RL cycle?
2. What will happen if PK is deficient?
3. What is the significance of 2,3-BPG ?



PK Deficiency in RBCs

- Through **bypassing** the **pyruvate kinase** step **2 ATP** were lost so the net energy yield in RBCs becomes 0.

- Lack of ATP →→ disturbance of cation gradient across the erythrocytic cell membrane →→Na & K are kept inside RBCs →→Na conc. outside RBCs is higher →→ loss of water →→ cell dehydration →→ premature destruction of the erythrocyte (hemolysis mainly in spleen).

4. What are the general characteristics of hemolytic anemia?

1. Pallor and jaundice.
2. Normocytic anemia.
3. High bilirubin (mostly unconjugated).
4. ↑Reticulocytic count

5. What are the different types of hemolytic anemia?

I-Hemolytic anemia caused by enzyme deficiency

- Pyruvate kinase deficiency hemolytic anemia

- Glucose-6-phosphate dehydrogenase (G6PD) deficiency hemolytic anemia (Favism)

II-Hemolytic anemia caused by abnormalities in Hb

- **Sickle cell anemia**

  A mutation occurs in βchain:

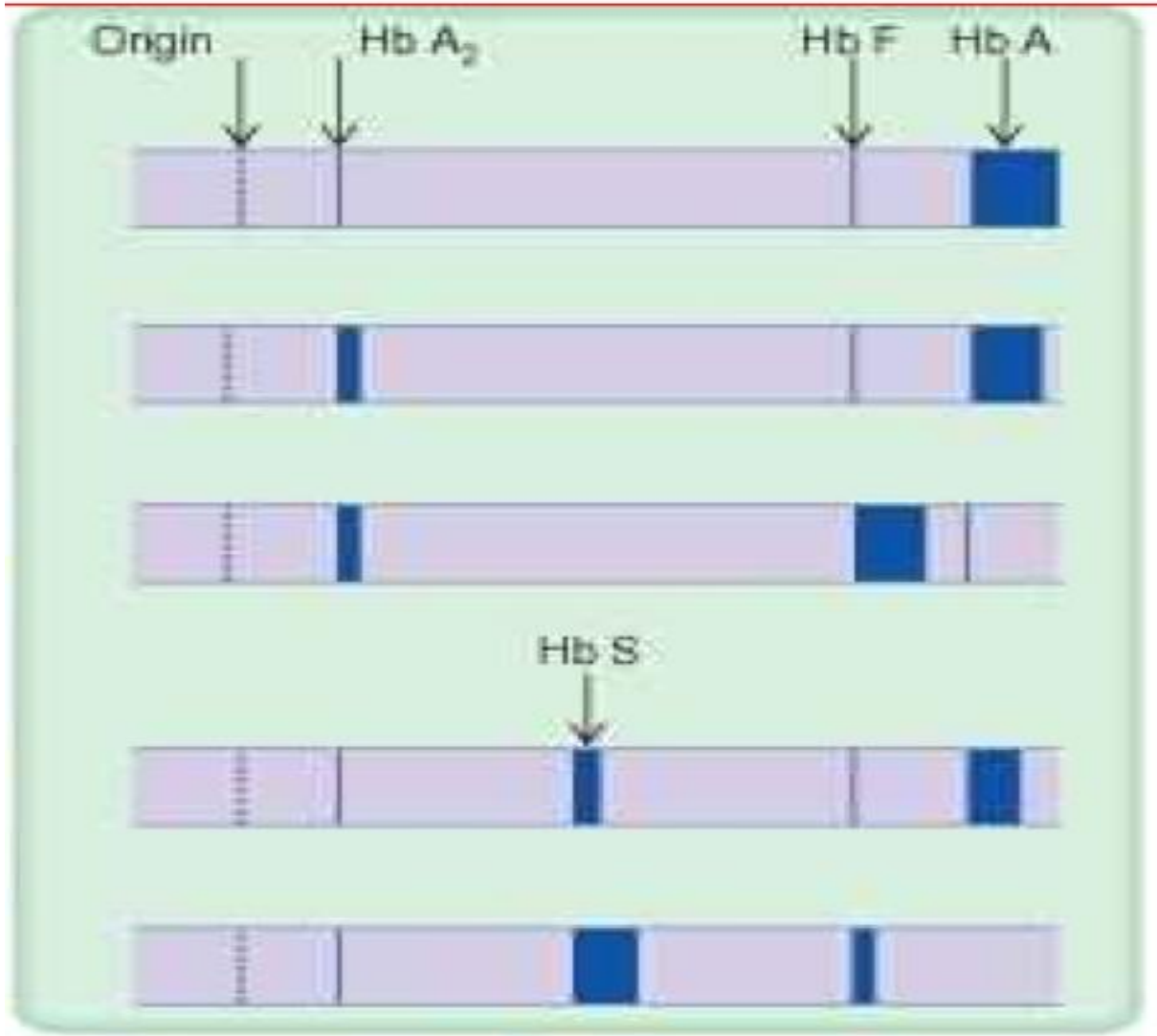
  Glutamic acid substituted by Valine

- **Thalassemia**

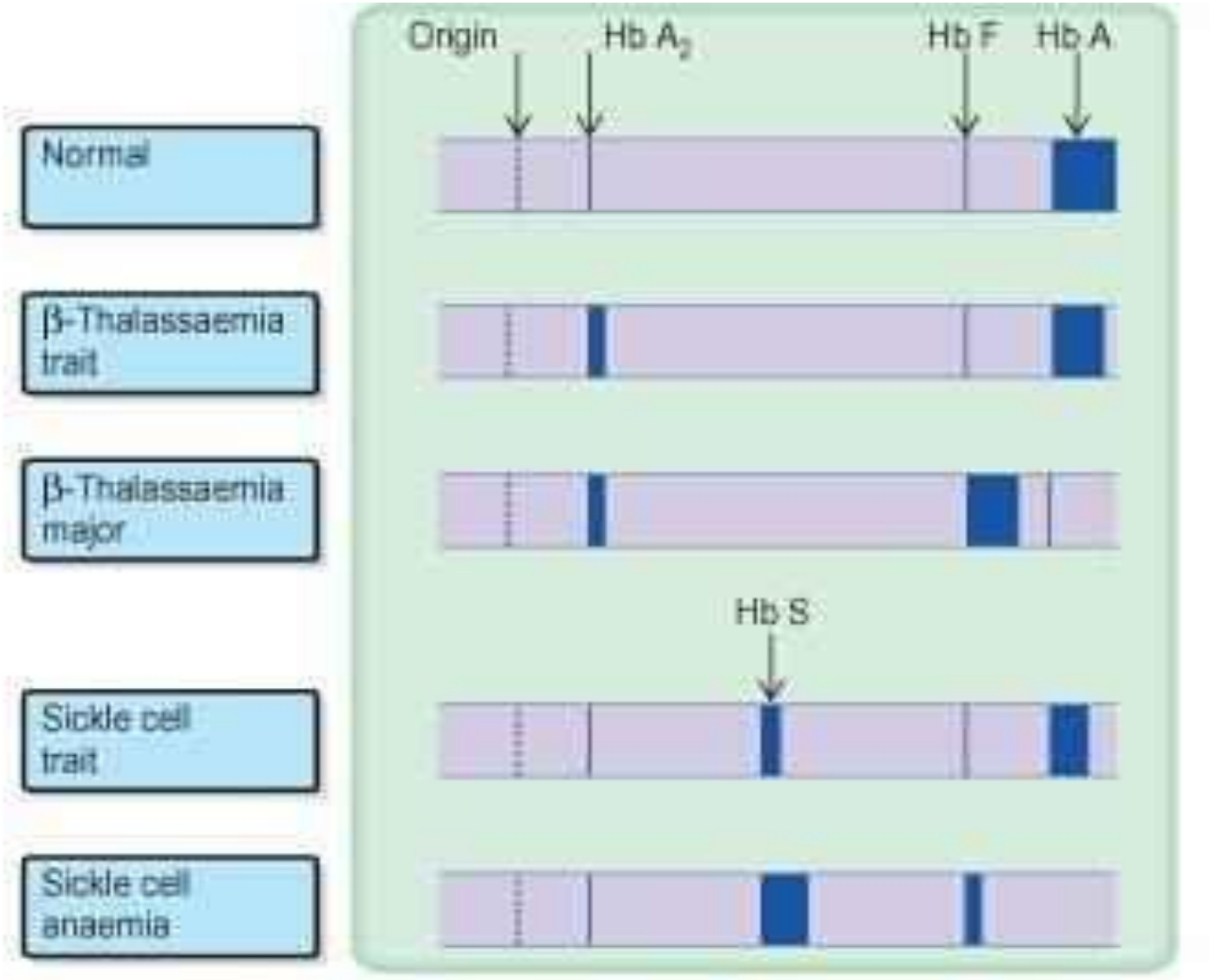
  Decreased rate of synthesis of globin chains

6-What is the Diagnostic test for anemia caused by abnormalities in Hb?

Hb electrophoresis



Hb Electrophoresis



7. What is the role of G6PD in RBCs?

Glucose-6-phosphate dehydrogenase

The rate-limiting enzyme of hexose monophosphate pathway (pentose phosphate pathway).

What is the importance of HMP Pathway?

- **Pentoses**
  - Nucleotides
  - DNA, RNA, ATP, NAD, FAD etc
- **NADPH**
  - Synthetic pathways: FA, Steroids
  - Antioxidant coenzyme (GSSG reductase)
  - Cytochrome P450 coenzyme
  - Bactericidal action
  - Synthesis of nitric oxide

G6PD

- **Is the main NADPH SOURCE**

Which Is Essential For The

Maintenance Of The G-SH Pool

(Coenzyme of G-SH reductase)

Role Of G-SH

1. **Protects RBCs membrane lipids from oxidation**

Oxidation → (hemolysis).

2. **Maintain the reduced states of Hb ferrous iron** to prevent accumulation of methemoglobin

3. **Protects globin from**

Oxidation → formation of denatured proteins that form insoluble masses that attach to the cell membranes (Heinz bodies)

4- Protect against Oxidation of membrane proteins

that causes RBCs to be rigid (less deformable), and removed from the circulation by macrophages in the spleen and liver.

8. Explain the presence of neurological symptoms in PKD anemia.

Prolonged hemolysis and heme breakdown →→ High levels of bilirubin →→ Indirect bilirubin crosses blood brain barrier →→ Deposition of bilirubin in the brain, e.g., basal ganglia →→ neurological symptoms (hypotonia and lethargy)

N.B.:

If the patient wasn't adequately treated, permanent neurological damage could have happened.

9. Explain the occurrence of hyperbilirubinemia.

Pyruvate kinase deficiency → Impaired glycolysis in RBCs with no ATP production →→ Loss of Na/K pump activity →→ Dehydration of RBCs →→ Hemolysis and heme breakdown →→ High levels of bilirubin (mainly indirect)

10. Explain the biochemical basis of phototherapy in hyperbilirubinemia.

Phototherapy:

Blue light converts bilirubin to a more polar isomer easily excreted in bile.



11-Mention different causes of deficiency anemia.

Deficiency anemia:

- It is caused by dietary deficiency of a factor needed for erythropoiesis:

(a) Iron deficiency anemia

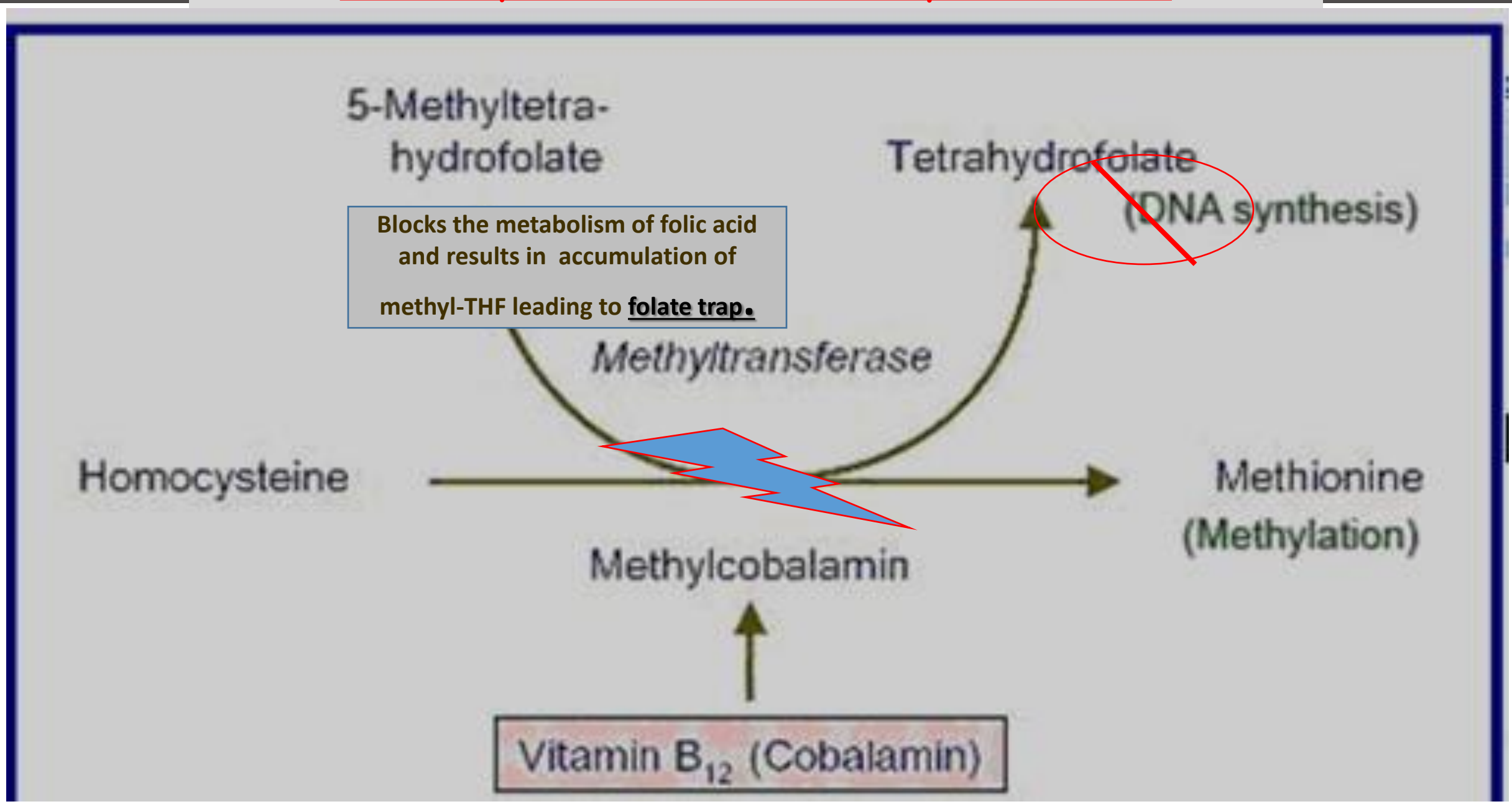
(b) Vitamin B12 or Folic acid deficiency:

- Deficiency of either vitamin b12 or folic acid or both causes **megaloblastic anemia (maturation failure anemia).**

Megaloblastic anemia

- Vitamin B_{12} deficiency blocks the metabolism of folic acid and results in accumulation of methyl-THF leading to folate trap. will lead to impaired purine and pyrimidine synthesis.
- This leads to Impaired DNA synthesis and prevention of cell division with the accumulation of immature erythrocytes (megaloblasts) in the circulation.

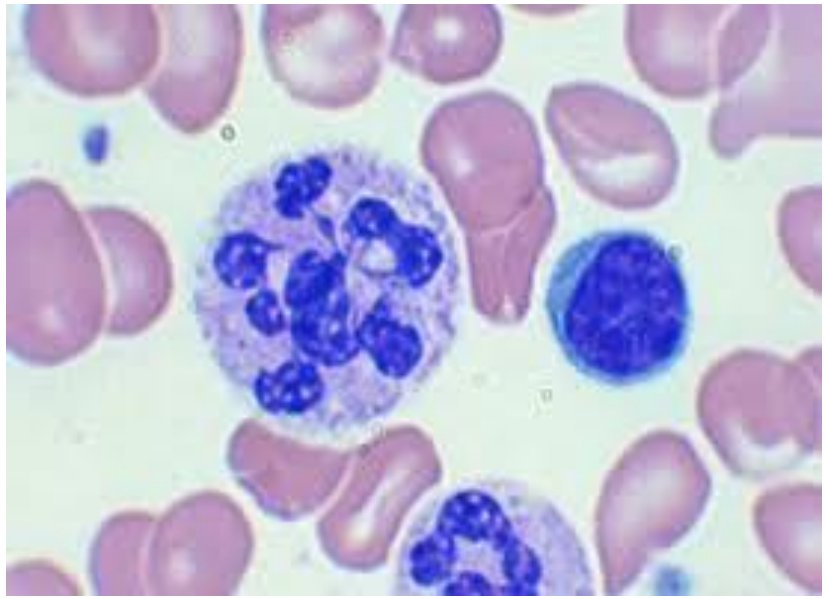
B12 is required for an essential enzymatic reaction



Folic acid Deficiency

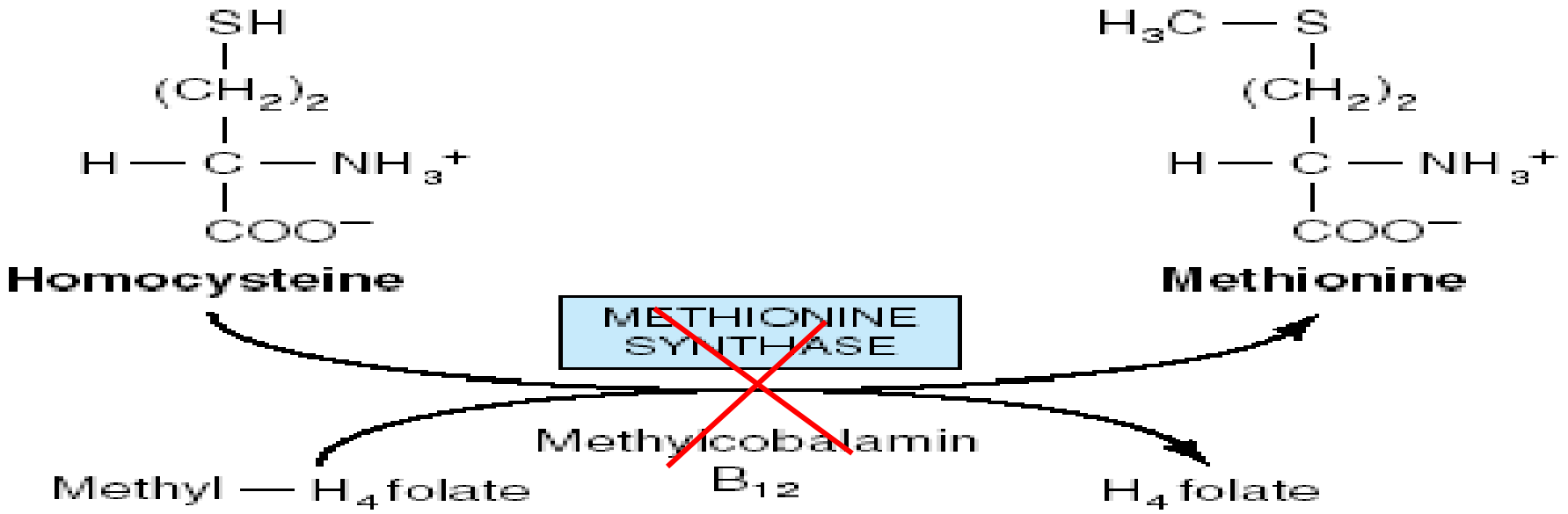
It may occur due to:

- Increased demand.
- Poor absorption.
- Vitamin B12 deficiency.
- Drugs.



Deficiency leads to Megaloblastic anemia due to accumulation of large immature RBC precursors (megaloblasts).

Folate Trap



In case of vit B12 deficiency the methylated form of THF is accumulated ,Because it can not be converted to other forms of folic acid so it is trapped in the methyl form .

So the levels of other forms of folic acid decrease , decreasing purine and thymine synthesis.

12-What are the unique characteristics of iron deficiency anemia?

- **Low Hb level**

Microcytic hypochromic anemia

- **Iron profile**

- Low serum iron level
- High Total iron binding capacity (TIBC)
- Low Ferritin levels

- **Microcytic anemia** means small RBCs.
- **Hypochromic** means that the red blood cells have less hemoglobin than normal.
- **Microcytic hypochromic anemia is due to:**

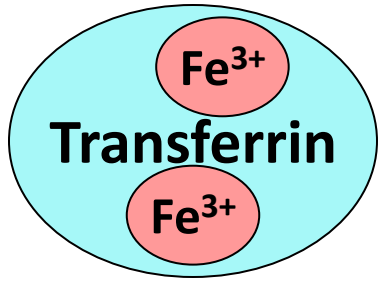
➢ The body has no enough iron available to incorporate into hemoglobin.

➢ Decresed ferrochelatase enzyme activity .

➢ Leads to inhibition of heme synthesis.

High Total iron binding capacity (TIBC):

- Represents the capacity of transferrin to bind iron.
- It measures the free transferrin that is ready to carry Iron (better than measuring transferrin itself).
- Transferrin transports iron (2 mol of Fe^{3+} per mol of Tf) in the circulation to sites where iron is required.



- Normally, the protein is only one third(35%) saturated with iron.
- High TIBC levels typically indicate low levels of iron in the blood.

Ferritin: is the predominant storage form of iron in cells.

- **Apoferritin** (the protein part of ferritin) can take 4000-4500 ferric ions / molecule to form ferritin.

4500 Fe3+ ions ferritin

- Ferritin is the primary intracellular iron-storage globular protein. It keeps iron in a soluble & non-toxic form.
- If plasma iron is decreased, iron in ferritin is transported easily to replenish transferrin in the plasma.
- So in iron deficiency anemia ferritin is markedly decreased.

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13-What's the daily iron requirement? What are iron sources?

- The average daily iron requirement is **15-20 mg/day**.
- **Iron sources:**

I-Heme iron (animal sources)

Found in; meat, poultry, fish.

Heme iron is easily absorbed and used by the body.

II-Non heme (inorganic) iron (plant sources)

Found in; dark green vegetables, fruits, grains, lentils, molasses and nuts.

Non heme iron is poorly absorbed by the body.

14-What are the factors affecting iron absorption and comment on its regulation?

Factors affecting iron absorption

1. **Requirements of body: An increase rate of erythropoiesis (e.g. after hemorrhage).**

2. **Vitamin C helps reduction of ferric to ferrous and forming soluble complex.**

3. **Gastric HCl helps absorption, so absorption decrease in achlorhydria and partial gastrectomy.**

4. **Some molecules in diet as tannic acid, phytate, phosphate and oxalates forming insoluble complexes with iron.**











*oxalic acid found in spinach

*phosphates found primarily in milk, dairy products and egg whites

*phytates in beans

*tannic acid in tea and coffee

form insoluble complexes with iron

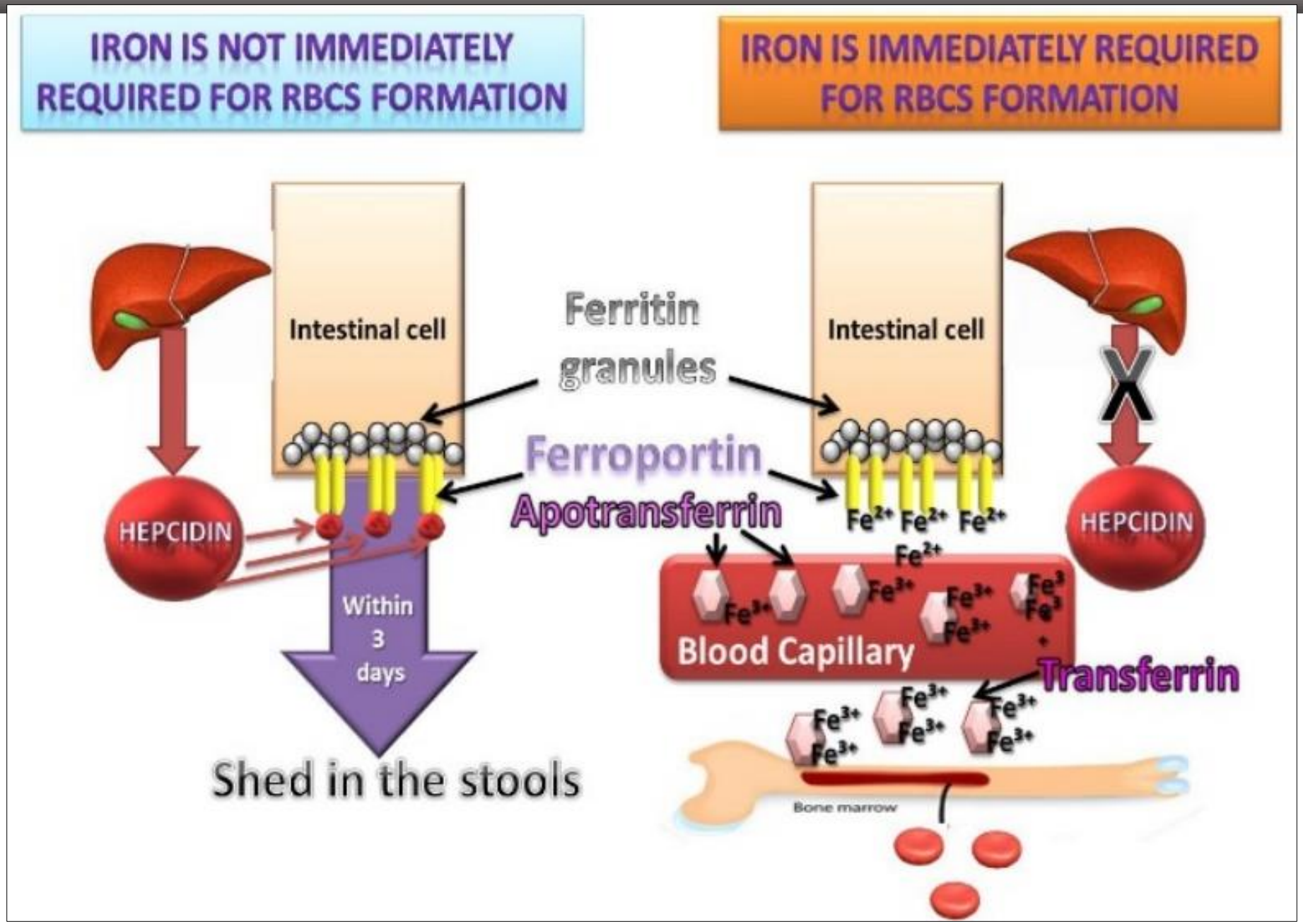
Vit C (Ascorbic acid)

Forms a soluble complex

Facilitating absorption and

helps reduction of Fe^{3+} to Fe^{2+}

Iron absorption regulation


IRON IS NOT IMMEDIATELY REQUIRED FOR RBCS FORMATION
IRON IS IMMEDIATELY REQUIRED FOR RBCS FORMATION
Intestinal cell
Ferritin granules
Ferroportin
Apotransferrin
HEPCIDIN
Within 3 days
Shed in the stools
Intestinal cell
X
HEPCIDIN
Fe2+
Blood Capillary
Transferrin
Fe3+
Bone marrow

Hepcidin

- **Ferroportin protein** (the only known exporter of iron from cells to blood) is regulated by the hepatic peptide ***hepcidin*** that induces internalization and lysosomal degradation of ferroportin.
- It is a hormone released from the liver when iron levels in the body becomes too high.

- It inhibits further iron passage from the small-intestinal epithelial cell into the blood by attaching and inhibiting the ferroportin transporter.

- A deficiency of hepcidin causes tissue iron overload.
- Hepcidin has a central role in iron homeostasis (its transcription is suppressed when iron is deficient).

So Total body Iron is regulated at the level of absorption by enterocytes of the duodenum

Mucosal Block



The intestinal content of ferritin is limited so once saturated with iron, the absorption of the iron is inhibited.

Mucosal Block

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Fe3+ ions
ferritin

When iron stores (ferritin), in the body are depleted absorption increases.

15-What are the specific treatment measures of iron deficiency anemia?

1. Diet Improvment :

- ✓ Eat sources of heme iron e.g; meat& fish.
- ✓ Avoid substances that inhibit iron absorption e.g; tea and coffee(taken 2 hrs after meal).

2. Oral iron supplementation:

- ✓ In case of vegan diet must be taken 2 hrs before meal.
- ✓ Follow up with CBC and iron profile.

3. Intravenous infusion of iron

- ✓ First line of ttt in severe cases Hb < 5 g/dl
- ✓ If no improvement to oral iron, may be recommended.


Thank You